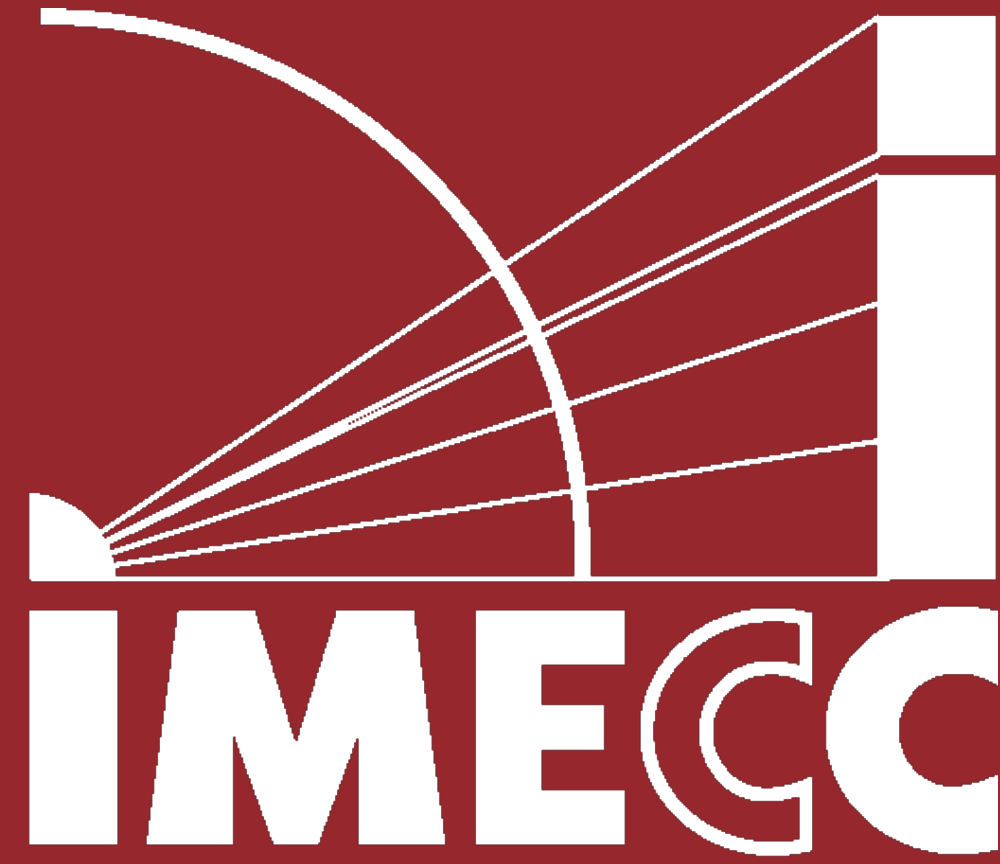


The Gauss class number problem via modular functions



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I conjecture $b(D) = 1$ iff $-D \in \{3, 4, 7, 8, 11, 12, 16, 19, 27, 28, 43, 67, 163\}$



What did Gauss conjecture?

Equivalent quadratic forms (positive, primitive)

$$g(x, y) = f(px + qy, rx + sy), \quad ps - qr = 1.$$

Discriminant of $ax^2 + bxy + cy^2$ is $D := b^2 - 4ac$. The set of equivalence classes $\text{Cl}(D)$ has cardinality $b(D)$, which is the **class number**.

Imaginary quadratic orders are subrings $\mathcal{O}$ of some imaginary quadratic field K which are free $\mathbb{Z}$ -modules of rank 2.

Fractional ideals of $\mathcal{O}$ are sets of the form $\alpha \mathfrak{a}$, where $\alpha \in K$ and $\mathfrak{a} \subset \mathcal{O}$ is an ideal. The invertible fractional ideals form a group $I(\mathcal{O})$, which modulo principal ideals is the class group

$$\text{Cl}(\mathcal{O}) = I(\mathcal{O})/P(\mathcal{O}), \quad \text{with cardinality } b(\mathcal{O}).$$

How do modular functions creep in?

The j -invariant is the function defined on the upper half-plane by

$$j(\tau) = 12^3 \frac{E_4(\tau)^3}{E_4(\tau)^3 - E_6(\tau)^2}.$$

Complex multiplication: $\tau \in \mathbb{H}$ is a CM-point if $\mathcal{O} = \{\alpha \in \mathbb{C} : \alpha(\mathbb{Z} + \mathbb{Z}\tau) \subset \mathbb{Z} + \mathbb{Z}\tau\}$ is larger than $\mathbb{Z}$. Then $\mathcal{O}$ is an order of $\mathbb{Q}(\tau)$ and $j(\tau)$ is an algebraic integer of degree $b(\mathcal{O})$.

How did Heegner prove it?

Weber showed that $\gamma_2(\tau) := j(\tau)^{1/3}$ also has degree $b(d_K)$, not $3b(d_K)$!

Thm. (Heegner-Stark) For an imaginary quadratic field $K = \mathbb{Q}(\sqrt{-d})$,
 $b(\mathcal{O}_K) = 1 \iff d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$.

Let $d_K = -p \equiv 5 \pmod{8}$ (the other cases are elementary). Using CM thrice, Heegner showed that the singular moduli

$$j(\sqrt{-p}) \quad \text{and} \quad \mathfrak{f}(\sqrt{-p})^2 \quad \text{have degree 3}$$

and that $\gamma_2(\tau_0) \in \mathbb{Z}$, for $\tau_0 := (3 + \sqrt{-p})/2$. Let $\alpha := e^{-\pi i/4} \mathfrak{f}_2(\tau_0)^2$. Rewriting α in terms of $\mathfrak{f}(\sqrt{-p})^2$ shows that it has degree 3. Now rewriting $\gamma_2(\tau_0)$ in terms of $\mathfrak{f}_2(\tau_0)$, we get

$$\gamma_2(\tau_0) = (\mathfrak{f}_2(\tau_0)^{24} + 16)/\mathfrak{f}_2(\tau_0)^8.$$

So $\alpha \in \mathcal{O}_K$ and the minimal polynomial of α^4 is $p(x) = x^3 - \gamma_2(\tau_0)x - 16$. Let's compute $p(x)$ in another way, by lifting the minimal polynomial $x^3 + ax^2 + bx + c$ of α to $p(x)$. This gives the Diophantine equation

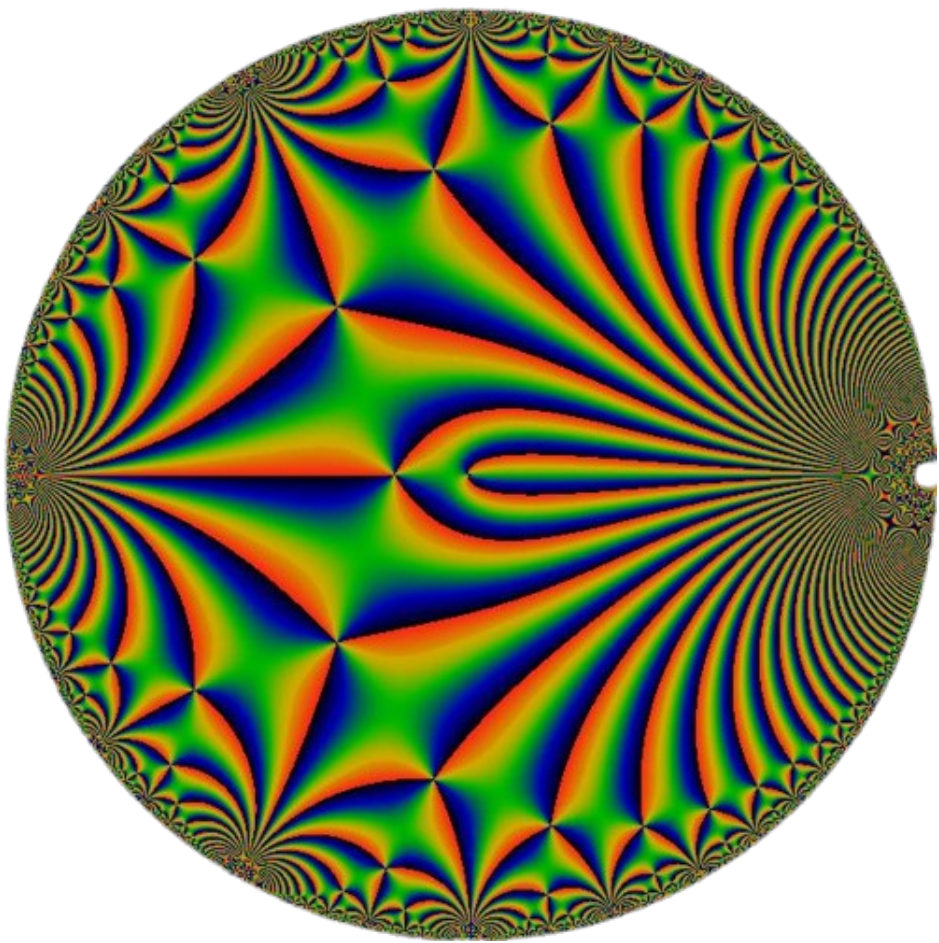
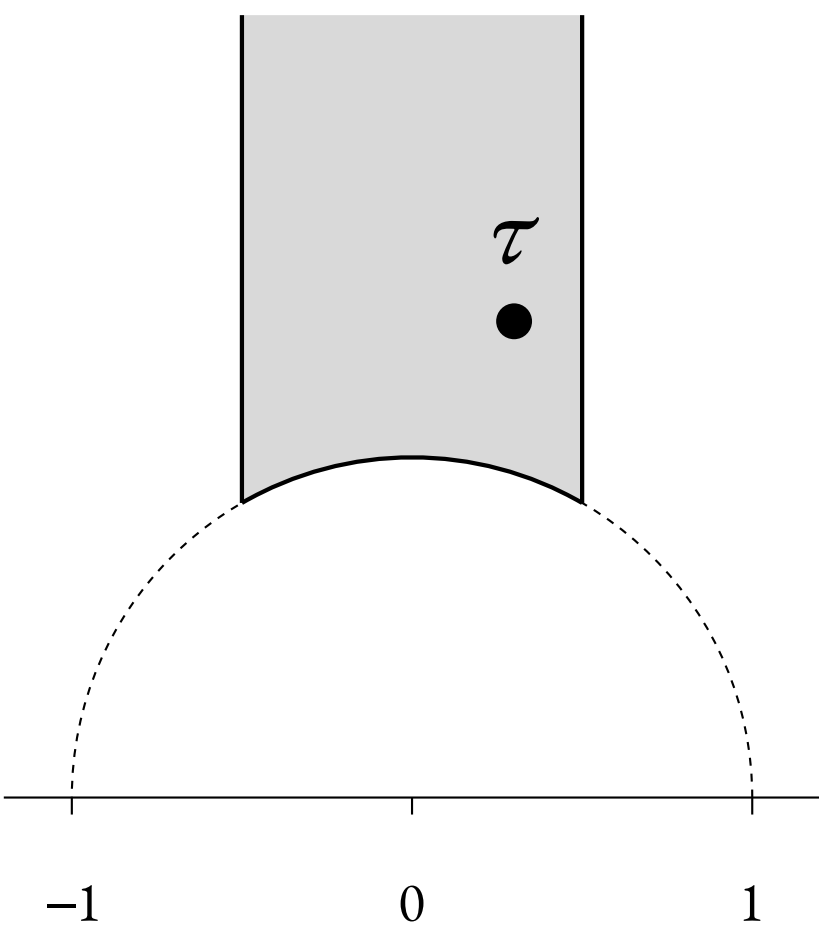
$$2X(X^3 + 1) = Y^2,$$

which has exactly **six solutions**. ■

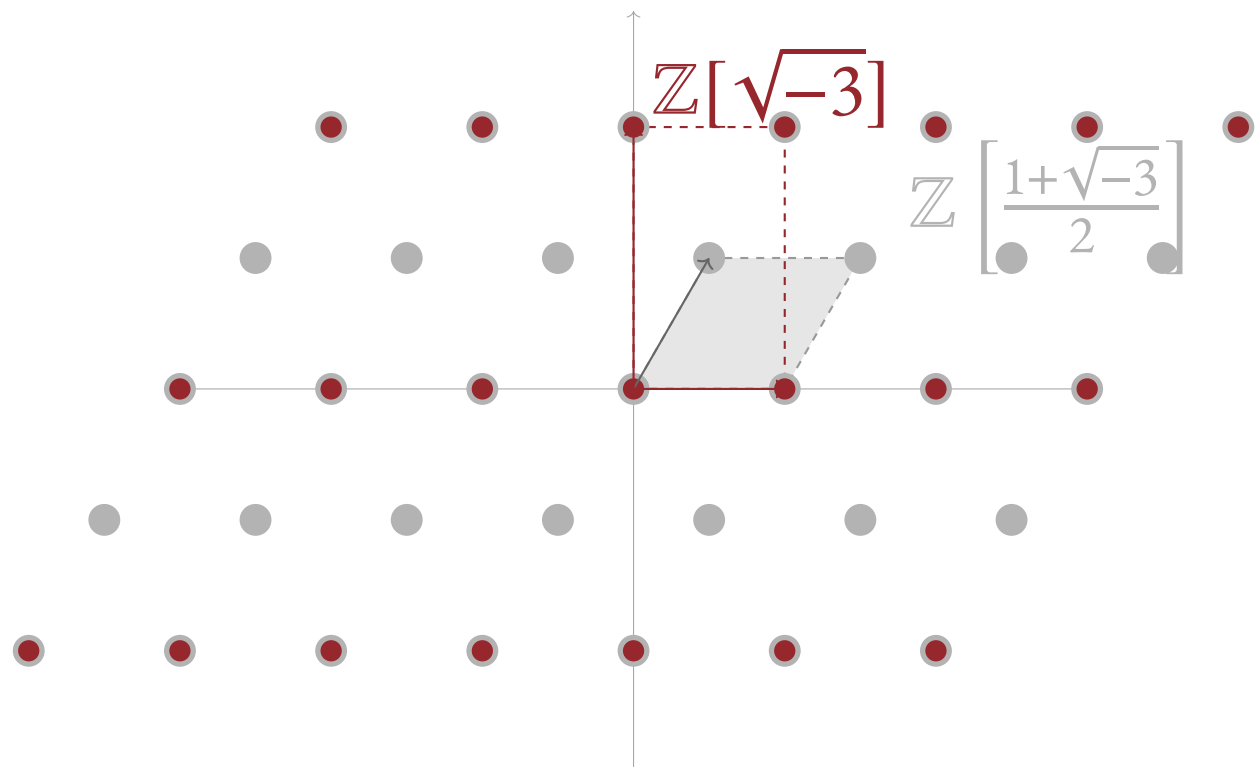
D	$b(D)$	Reduced Forms
-4	1	$x^2 + y^2$
-8	1	$x^2 + 2y^2$
-12	1	$x^2 + 3y^2$
-20	2	$x^2 + 5y^2, 2x^2 + 2xy + 3y^2$
-28	1	$x^2 + 7y^2$
-56	4	$x^2 + 14y^2, 2x^2 + 7y^2, 3x^2 \pm 2xy + 5y^2$

$$d_K = \begin{cases} 4d, & d \not\equiv 1 \pmod{4}, \\ d, & d \equiv 1 \pmod{4}. \end{cases}$$

$$\begin{aligned} \text{Cl}(D) &\xrightarrow{\sim} \text{Cl}(\mathcal{O}) \\ a|x - \tau y|^2 &\mapsto \mathbb{Z} + \mathbb{Z}\tau \end{aligned}$$

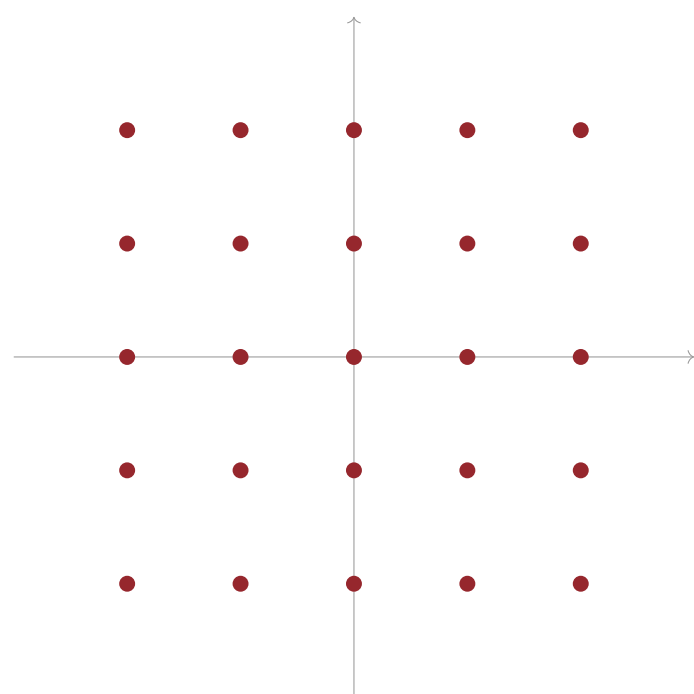


I proved it borrowing some magical algebra from Weber.

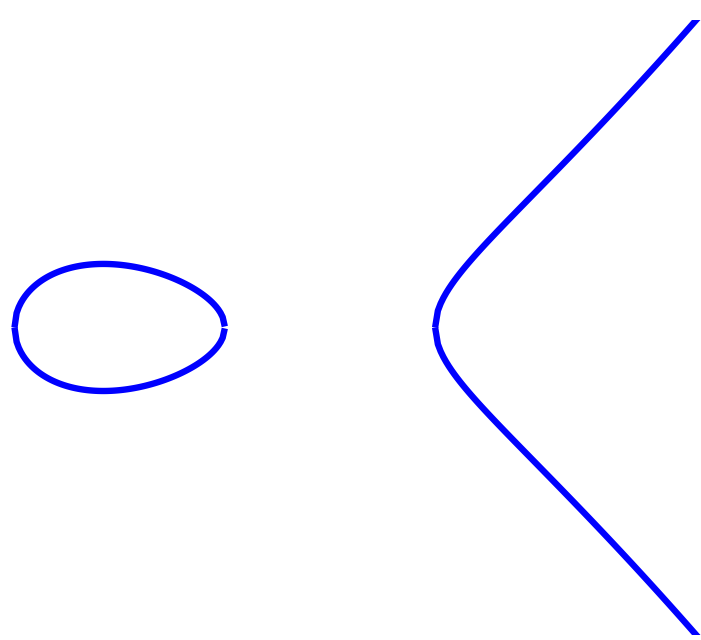


Weber functions

$$\begin{aligned} \mathfrak{f}(\tau) &= e^{-\pi i/24} \eta((\tau+1)/2)/\eta(\tau), \\ \mathfrak{f}_1(\tau) &= \eta(\tau/2)/\eta(\tau), \\ \mathfrak{f}_2(\tau) &= \sqrt{2} \eta(2\tau)/\eta(\tau). \end{aligned}$$



$\mathbb{H}$



(Full report.)